MySQL 5.7 Reference Manual

Version 5.7 Features and Functionality

1. Authentication and User Management

MySQL 5.7 uses the mysql_native_password as the default authentication plugin. The syntax for creating users is:

```sql
CREATE USER 'username'@'host' IDENTIFIED BY 'password';
```

Password validation can be enforced through the validate_password plugin, which must be installed separately:

```sql
INSTALL PLUGIN validate_password SONAME 'validate_password.so';
```

2. Important System Variables

Variable Name	Default	Description
max_connections	151	Maximum permitted number of client connections
innodb_buffer_pool_size	128M	Size of the buffer pool for caching data and indexes
query_cache_size	1M	Size of memory allocated for caching query results
query_cache_type	ON	Query cache type (ON, OFF, DEMAND)

The query cache is still supported but will be removed in future versions.

3. SQL Syntax Elements

GROUP BY Handling

In MySQL 5.7, ONLY_FULL_GROUP_BY SQL mode is enabled by default, which requires all non-aggregated columns in the SELECT list to appear in the GROUP BY clause.

JSON Support

MySQL 5.7 introduces JSON data type with functions for JSON manipulation:

```sql
CREATE TABLE products (
  id INT PRIMARY KEY,
  attributes JSON
);

INSERT INTO products VALUES (1, '{"color": "red", "size": "medium"}');

SELECT JSON_EXTRACT(attributes, '$.color') FROM products;
```

4. Deprecated Features

The following features are deprecated in MySQL 5.7 and may be removed in future versions:

- ENCRYPT() function
- PASSWORD() function
- OLD_PASSWORD() function
- The query cache
- mysql_install_db script (replaced by mysqld --initialize)
- TIMESTAMP data type automatic initialization and updating properties
- Spatial functions with prefixes (use unprefixed versions instead)
- YEAR(2) data type

5. InnoDB Enhancements

- InnoDB is the default storage engine
- InnoDB supports full-text indexes
- The InnoDB buffer pool can be resized dynamically
- InnoDB supports native partitioning

6. Error Codes

Common error codes in MySQL 5.7:

- 1046: No database selected
- 1062: Duplicate entry for key
- 1064: Syntax error
- 1146: Table doesn't exist
- 1215: Cannot add foreign key constraint
- 1451: Cannot delete or update a parent row (foreign key constraint)

7. Replication

MySQL 5.7 supports:

- Multi-source replication
- Row-based, statement-based, and mixed replication formats
- Global transaction identifiers (GTIDs)
- Semi-synchronous replication
- Delayed replication

The default replication format is still STATEMENT.